

Stakeholder Analyses of Firm-Related Web Forums: Applications in Stock Return Prediction

DAVID ZIMBRA, Santa Clara University

HSINCHUN CHEN and ROBERT F. LUSCH, University of Arizona

In this study, we present stakeholder analyses of firm-related web forums. Prior analyses of firm-related forums have considered all participants in the aggregate, failing to recognize the potential for diversity within the populations. However, distinctive groups of forum participants may represent various interests and stakes in a firm worthy of consideration. To perform the stakeholder analyses, the Stakeholder Analyzer system for firm-related web forums is developed following the design science paradigm of information systems research. The design of the system and its approach to stakeholder analysis is guided by two kernel theories, the stakeholder theory of the firm and the systemic functional linguistic theory. A stakeholder analysis identifies distinctive groups of forum participants with shared characteristics expressed in discussion and evaluates their specific opinions and interests in the firm. Stakeholder analyses are performed in six major firm-related forums hosted on Yahoo Finance over a 3-month period. The relationships between measures extracted from the forums and subsequent daily firm stock returns are examined using multiple linear regression models, revealing statistically significant indicators of firm stock returns in the discussions of the stakeholder groups of each firm with stakeholder-model-adjusted R^2 values reaching 0.83. Daily stock return prediction is also performed for 31 trading days, and stakeholder models correctly predicted the direction of return on 67% of trading days and generated an impressive 17% return in simulated trading of the six firm stocks. These evaluations demonstrate that the stakeholder analyses provided more refined assessments of the firm-related forums, yielding measures at the stakeholder group level that better explain and predict daily firm stock returns than aggregate forum-level information.

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Authors' addresses: D. Zimbra (corresponding author), Operations Management and Information Systems, Leavey School of Business, Santa Clara University, 500 El Camino Real, Santa Clara, CA, 95053, USA; email: dzimbra@scu.edu; H. Chen, Artificial Intelligence Lab, Management Information Systems, Eller College of Management, University of Arizona, 1130 East Helen St, Tucson, AZ, 85721, USA; email: hchen@eller.arizona.edu; R. F. Lusch, Eller College of Management, University of Arizona, 1130 East Helen St, Tucson, AZ, 85721, USA; email: rlusch@eller.arizona.edu.

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1. INTRODUCTION

Web-based social media, including forums, reviews, and blogs, offers valuable business intelligence to firms. Social media platforms allow diverse constituencies to discuss various aspects of the firm, including its products and services, public and labor policies, and investor expectations. The social nature of these discussions provides rich information on the firm, revealing natural expressions and perspectives inaccessible through traditional market research [Kozinets 2002]. Discussions within social networks can also have great influence over the opinions of participants, even more so than traditional media [Watts and Dodd 2007; Goh et al. 2013]. Recognizing the importance of these feedback channels, researchers have advocated for the development of approaches for the analysis of social media [Godes et al. 2005; Chen 2010].

The information conveyed in firm-related social media has been used to explain various indicators of firm performance, including product sales and stock behavior. Studies typically focus on the volume and sentiment (valence or opinion) of social media communications. The relationship between book reviews and book sales has been widely examined [Forman et al. 2008; Hu et al. 2008; Zhang et al. 2012]. Various social media have been applied to explain movie box office returns including movie reviews [Duan et al. 2008], forum discussions [Liu 2006], and Twitter messages [Rui et al. 2010]. Researchers have also utilized the discussions in firm-related web forums to explain and predict stock behavior [Antweiler and Frank 2004; Das and Chen 2007]. In these analyses, the communications of all participants discussing a particular product or firm are typically aggregated to form social media measures at the product or firm level. However, such an approach fails to recognize the potential for diversity in participants and their distinctive values and interests in the product or firm. Social media platforms are open communication environments freely available online and populated by a wide variety of participants. Simply aggregating all participant communications could muddle the expressions of distinctive perspectives or miss minority viewpoints, diminishing the analytic effectiveness.

While prior studies have shown that firm-related web forums offered significant information in explaining firm stock behavior, they reported mixed results and difficulties in the analysis of observably noisy forum environments [Antweiler and Frank 2004; Das and Chen 2007]. Their analytic approaches viewed all forum participants uniformly, and these findings may be attributed to diversity within the forum and the complexities associated with evaluating diverse expressions on the firm. Distinctive groups of forum participants may represent various interests and stakes in a firm worthy of consideration. Therefore, in this study, we present stakeholder analyses of firm-related web forums. A stakeholder analysis identifies distinctive groups of forum participants with shared characteristics expressed in discussion and evaluates their specific opinions and interests in the firm. We perform stakeholder analyses in six major firm-related forums hosted on Yahoo Finance over a 3-month period. Following prior studies [Antweiler and Frank 2004; Das and Chen 2007], we examine the relationships between measures extracted from the forums and subsequent daily firm stock returns and apply them in daily stock return prediction. The results indicate that the stakeholder analyses provide more refined assessments of the firm-related forums, yielding distinctive stakeholder groups and measures that better explain and predict subsequent firm stock returns than aggregate forum-level information.

To perform the stakeholder analyses, the Stakeholder Analyzer system (SA) for firm-related web forums is developed following the design science paradigm of information systems research [Hevner et al. 2004]. The SA represents the instantiation of an information system designed to perform stakeholder analysis in firm-related forums using the guidance of two kernel theories, the stakeholder theory of the firm [Donaldson and Preston 1995] and the systemic functional linguistic theory [Halliday 2004]. The SA is

composed of many integrated technical components, some adopted from prior research, each carefully selected based on the kernel theories and problem characteristics. We describe and justify the design of the SA but recognize that design alternatives exist, and the design process is an inherently iterative search [Hevner et al. 2004]. The results of the stakeholder analyses and stock return prediction demonstrate the SA is effective in forum analysis, validating its design and also promoting future design iterations.

2. LITERATURE REVIEW

The foundation for this research is established through the review of several areas of the literature. First, a brief review of the literature on stock behavior and prediction is provided, followed by review of the literature examining the relationship between firm-related web forum discussions and stock behavior. The design of the stakeholder analysis approach was inspired and guided by two kernel theories, reviewed next: the stakeholder theory of the firm and the systemic functional linguistic theory. A stakeholder analysis recognizes that various stakeholder groups within the firm-related forums may have distinctive interests in the firm and opinions worthy of consideration. Stakeholder groups are established based on shared characteristics and interests in the firm expressed in forum discussion. Therefore, quantifiable features characterizing their participation in forum discussion are required. The systemic functional linguistic theory offers a comprehensive representation of the various functions fulfilled by language and communication. This kernel theory guides the definition of a comprehensive feature representation of forum participation to serve as the basis for a stakeholder analysis.

2.1. Stock Behavior and Prediction

In the literature on stock behavior, there are two prominent theoretical perspectives, the efficient market hypothesis and the random walk theory. According to the efficient market hypothesis [Fama 1964], at any point in time the price of a stock reflects all available information. Since the market reacts instantaneously to new information, it is impossible to consistently outperform. The random walk theory [Malkiel 1973] suggests that the price of a stock varies randomly, making outperforming the market through prediction impossible. Despite these pessimistic theoretical perspectives, the predictability of stock behavior has been empirically demonstrated [Lo and MacKinlay 1988; Jaffe et al. 1989; Pesaran and Timmermann 1995].

Stock prediction has followed two general approaches, the fundamentalist and the technician. According to the fundamentalist, the price of a stock is determined by the fundamental and financial measures of the economy, industry, and firm, including economy and sector indicators and financial ratios of the firm [Brownstone 1996; Atiya et al. 1997; Cao et al. 2005]. The technician approach utilizes historical time-series information of the stock and market behavior, including price, volatility, and trading volume trends, to predict future returns [Barnes et al. 2000; Pai and Lin 2005; Hadavandi et al. 2010; Ticknor 2013]. In simulated trading, fundamentalist strategies correspond to longer waiting times before reacting to information on the firm, while technicians respond quickly to capture profits before the market fully absorbs the information into the stock price [LeBaron et al. 1999].

Researchers have also evaluated additional sources of information on the firm that have been shown to provide powerful indications of future stock behavior. Studies have examined the numeric and textual content of financial reports and disclosures issued by the firm, such as 10-Ks, and found them to contain valuable information on upcoming firm performance [Kothari et al. 2009; Campbell et al. 2011; Lehavy et al. 2011; Lin et al. 2011; Kravet and Muslu 2013]. Professional news articles on the firm from the business press have also offered significant indications of subsequent stock behavior [Tetlock 2007; Kothari et al. 2009]. To capitalize on the information released through the press

quickly before other investors can react, researchers have incorporated the content of news articles into their stock prediction models [Lavrenko et al. 2000; Mittermayer 2004; Schumaker and Chen 2009a, 2009b; Schumaker et al. 2012].

2.2. Firm-Related Web Forums and Stock Behavior

Similarly, researchers have examined the discussions held in firm-related web forums and identified significant relationships with concurrent and subsequent firm stock behavior [Antweiler and Frank 2004; Das and Chen 2007; Chen et al. 2013]. The forums typically studied are hosted on major web portals like Yahoo Finance and intended to attract investors and discussion on stock-related topics. While early research focused on simple forum activity measures, such as the number of messages posted, methods to assess participant opinions expressed in discussion were later devised. These “bullishness” sentiment classifiers were designed to distinguish buy, hold, or sell investment positions [Antweiler and Frank 2004; Das and Chen 2007] but achieved only modest results attributed in part to an observably noisy discussion environment and the complexities of forum analysis [Das and Chen 2007]. Researchers have developed approaches to classify messages as signal or noise to improve the quality of information extracted from forums [Gu et al. 2007]. The difficulties in forum analysis have caused other researchers examining social media and stock behavior to avoid these sources altogether [Tirunillai and Tellis 2012; Luo et al. 2013]. However, sentiments expressed in forum discussions have provided valuable indications of subsequent stock prices [Das and Chen 2007].

Prior studies of firm-related forum discussions and stock behavior have several notable shortcomings, due to limitations in their analytic approaches, addressed in this study. The existing analytic approaches have viewed all participants in the forums uniformly as investors, reflected in the measures extracted (at the forum/firm level of aggregation) and the sentiment classification approaches designed to distinguish investment positions. Firm-related forums, particularly those open to the public and hosted on major web portals like Yahoo, are general and populated by a wide variety of participants with distinctive interests in the firm. The stakeholder analyses presented in this study identify distinctive stakeholder groups of participants with shared characteristics expressed in forum discussion and independently evaluates their specific opinions and interests in the firm. Additionally, the existing studies have been static and retrospective, seeking to establish past statistical relationships between forum discussions and stock behavior without applying these measures in true stock return prediction. We evaluate the relationships between measures extracted from the forums and subsequent firm stock return following prior studies [Antweiler and Frank 2004; Das and Chen 2007] and also apply them in daily stock return prediction and simulated trading of firm stock for further validation.

2.3. Stakeholder Theory of the Firm

Distinctive groups of forum participants with shared characteristics and interests expressed in discussion may represent diverse stakes in a firm worthy of consideration. Therefore, we perform stakeholder analyses of the firm-related web forum to evaluate these distinctive groups and use the stakeholder theory of the firm as a kernel theory to guide the design of the SA and its approach to stakeholder analysis. The stakeholder theory of the firm describes the diverse relationships the firm holds with its various constituency groups [Donaldson and Preston 1995]. Traditional, functional role-based stakeholder groups included investors, customers, communities, activists, employees, governments, political groups, and suppliers. Prior models suggested that a successful firm focuses on satisfying its shareholders and their interests. However, according to the stakeholder theory, the firm must actually be attentive to a diverse collection of

stakeholder groups, not just its shareholders, for long-term success [Donaldson and Preston 1995]. Empirical studies have shown that stakeholder management relates to improved financial performance [Berman et al. 1999; Harrison and Freeman 1999; Hillman and Keim 2001], and strategic management and marketing researchers have advocated its adoption [Freeman and McVea 2001; Lusch and Webster 2011; Webster and Lusch 2013].

Stakeholders of a firm are defined broadly, beyond simply functional role-based groups, as “any group or individual who can affect or is affected by the achievement of the organization’s objectives” [Freeman 1984, p. 46]. Stakeholders are identified “by their interest in the corporation, regardless of whether the organization has any corresponding functional interest in them” [Preble 2005, pp. 409–410]. Therefore, participants in firm-related web forums are by definition stakeholders of the firm. We use these terms interchangeably throughout this research. Through their engagement in discussion, participants act upon their interest in the firm and its activities. Forum discussion also influences the perspectives of other participants and shapes their opinions of the firm, ultimately impacting the ability of the firm to achieve its objectives.

In a stakeholder analysis, stakeholders are typically grouped based on underlying similarity with respect to the firm, so their shared concerns and interests can be evaluated and addressed. While stakeholder group membership was traditionally assigned based on functional role with the firm (investor, customer, employee, etc.), research has suggested that grouping stakeholders based on shared interests, priorities, or value preferences with the firm provides clearer and more actionable intelligence on key firm-related issues [Wolfe and Putler 2002; Payne et al. 2005]. Even within a role-based stakeholder group, such as investors or consumers, there may be great heterogeneity in interests or perspectives on firm-related issues. “Subgroups within different role-based stakeholder groups might have more similar priorities than either subgroup has with others within their role-based stakeholder group” [Wolfe and Putler 2002, p. 64]. Researchers have raised concerns that normative role-based stakeholder models are overly simplistic and do “not adequately represent the complex social and organizational realities managers face” [Gioia 1999, p. 228]. Therefore, to avoid the normative constraints of generic role-based stakeholder classification models and allow for the emergence of stakeholder groups based on shared interests, unsupervised clustering approaches have been successfully applied in stakeholder group identification and assignment [Wolfe and Putler 2002].

The insights presented in the stakeholder theory literature guide the design of the SA and its approach to stakeholder analysis. A stakeholder analysis groups forum participants using unsupervised clustering based on shared interests and characteristics expressed in forum discussion, rather than attempting to classify them by functional role. An unsupervised approach to stakeholder analysis closely follows the suggestions of the contemporary stakeholder analysis literature and does not presume functional role-based stakeholder groups in advance, but rather allows interest-based stakeholder groups to emerge without prejudice, consisting of participants expressing similar perspectives on key firm-related issues based on their forum activity. An alternative design, to develop stakeholder group classification models for supervised stakeholder analysis, would require the difficult stakeholder group identification task that has challenged firms and researchers [Payne et al. 2005], likely result in generic functional role-based stakeholder groups that provide inconsistent or contradictory perspectives on firm-related issues [Gioia 1999; Wolfe and Putler 2002; Payne et al. 2005], and inhibit the discovery of previously unknown stakeholder groups. Therefore, the stakeholder analyses are not expected to identify groups corresponding to functional role (investor, customer, employee, etc.), but rather identify groups of participants with shared interests, characteristics, and perspectives.

Stakeholder theory and analysis has also been utilized to inform the design and development of information systems [Pouloudi 1999; Mishra and Mishra 2013]. Prior studies typically focus on evaluating the prospective users of a proposed information system through traditional stakeholder analysis approaches. For example, existing research has examined stakeholders in the implementations of ERP systems [Boonstra 2006], telecommunications centers [Bailur 2007], and e-government systems [Zhang et al. 2005]. Frameworks for conducting stakeholder analysis in information systems development projects in health care [LaPointe et al. 2011] and e-government [Flak et al. 2005] have also been proposed.

Researchers have also created information systems and tools to support stakeholder analysis. Rowley [2011] developed a benefits analysis tool for understanding and aligning the interests of stakeholder groups in e-government. Lim et al. [2005] devised a system to support the strategic management of competing stakeholders using case-based reasoning and strategies derived from similar cases in the past. Chung et al. [2009] developed the business stakeholder analyzer system to assist management in the analysis of stakeholders of their firm on the web. The system automatically classified web pages that link to a firm's website into predefined role-based stakeholder groups to facilitate manual analysis. To our knowledge, the SA presented in this study is the first system designed and instantiated to perform stakeholder analysis in firm-related social media, although systems for this task have been requested in the literature [Sedereviciute and Valentini 2011].

2.4. Systemic Functional Linguistic Theory

To perform a stakeholder analysis and evaluate the shared characteristics and interests of participants expressed in discussion, quantifiable features characterizing their forum activity are required. The systemic functional linguistic theory (SFLT) offers a comprehensive representation of the various functions fulfilled by language and communication and is used as a kernel theory to guide the definition of a feature representation of forum participation to serve as the basis for the stakeholder analysis. Grounded in functional linguistics, the SFLT describes language as a social resource used to express meaning in context, composed of three meta-functions: ideational, interpersonal, and textual [Halliday 2004]. The ideational meta-function describes how language is used to convey ideas of the mental world. The interpersonal meta-function describes how communication serves as a mechanism for social interaction. And the textual meta-function describes how writing contains physical expressions of style and structure. These three SFLT meta-functions guide the definition of a comprehensive stakeholder feature representation to assess the various functions of language and communication fulfilled by forum discussion. We carefully adopt features from prior literature in topical analysis, social network analysis, and stylometric analysis that are aligned with each SFLT meta-function. We describe and justify the design of our stakeholder feature representation and aim to accurately and comprehensively represent the SFLT meta-functions, but recognize that design alternatives and other relevant features may exist and encourage future design iterations and research in the area.

Corresponding to the ideational meta-function of SFLT, content-based features are included to represent the ideas described by stakeholders in forum discussion. Stakeholder analysis and identifying stakeholder groups united based on shared interests, priorities, or value preferences with the firm may provide clearer and more actionable intelligence on key issues [Wolfe and Putler 2002; Payne et al. 2005]. These mutual interests are communicated in the similar ideas offered in discussion and embodied in the word content of messages. Content-based representations are often applied in the topical analysis of a document collection. In web forums, content-based features have been utilized to cluster discussion threads by topic [Wu and Li 2007; Wu et al.

2008; Yang and Ng 2008]. The most widely used content-based representation is a bag-of-words, or word unigram model, often extended to word n -grams.

Corresponding to the interpersonal meta-function of SFLT, social-network-based features are included to represent the interpersonal connections formed between stakeholders through forum discussion. A social network approach to stakeholder analysis is widely supported in the stakeholder literature [Rowley 1997; Pouloudi and Whitley 1997; Neville and Menguc 2006], and social network analysis techniques have been applied in social media communities to identify clusters of closely connected participants [Backstrom et al. 2006; Mishra et al. 2007]. Since the stakeholder analysis is performed to identify groups of similar forum participants with shared characteristics, interests, and perspectives on the firm, social-network-based features that measure the tendency for participants to interact with one another in discussion are adopted for the stakeholder representation. We specifically focus on these social-network-based features to leverage the expected participant behavior in an open forum environment that allows them to interact and develop relationships with any other members freely, simply by joining discussion threads. Confirmation bias [Plous 1993] suggests that forum participants will gravitate toward interaction with others who express similar interests and perspectives on firm-related issues as their own, to confirm their held hypotheses. The principle of homophily in sociology, the similarity of two members in a network, drives the creation and strength of ties in various social network types [McPherson et al 2001]. The homophily of forum participants may be revealed over time as they seek out others in the forum who are similar to them and engage them more frequently in discussion on the firm. Therefore, the social-network-based features adopted for the stakeholder representation measure the repeated interactions of participants in forum discussion threads to capture this behavior and affinity. Other measures from the social network analysis literature such as power, degree, or rank were not included in the design of the stakeholder feature representation since they focus on the prominence of individual nodes within a network and not directly on this behavior. These measures may also be more informative in social networks and social media platforms where ties are sparse or more difficult to form. However, in future studies, we intend to also evaluate the prominent stakeholders within forums.

Corresponding to the textual meta-function of SFLT, writing-style-based features are also included to represent the expressions of style in the writing of stakeholders. Densely connected stakeholder networks, such as those within forums, can produce strongly unified stakeholder groups [Rowley 1997]. The bonds between stakeholders of a particular group may be expressed socially through the adoption of similar communication practices, exhibited in their writing style. Social psychologists have revealed a strong de-individuation effect in web forums, where participants alter their individual communication behaviors to conform to their group norms [Spears et al. 2002]. Research from stylometric analysis has also shown that authors with similar background, training, and interests are characterized by similar writing styles [Baayen et al. 2002]. The writing-style-based features adopted for the stakeholder representation correspond to the stylometric feature representations developed in prior studies for authorship attribution [Zheng et al. 2006; Abbasi and Chen 2008a, 2008b].

3. STAKEHOLDER ANALYZER SYSTEM FOR FIRM-RELATED WEB FORUMS

To perform stakeholder analyses in firm-related web forums, the SA system is developed consisting of three major components, web forum collection, sentiment analysis, and stakeholder analysis, as depicted in Figure 1. The SA represents the instantiation of an information system following the design science paradigm of information systems research [Hevner et al. 2004]. The design of the SA and its approach to stakeholder analysis closely follows the guidance of the two kernel theories. The stakeholder

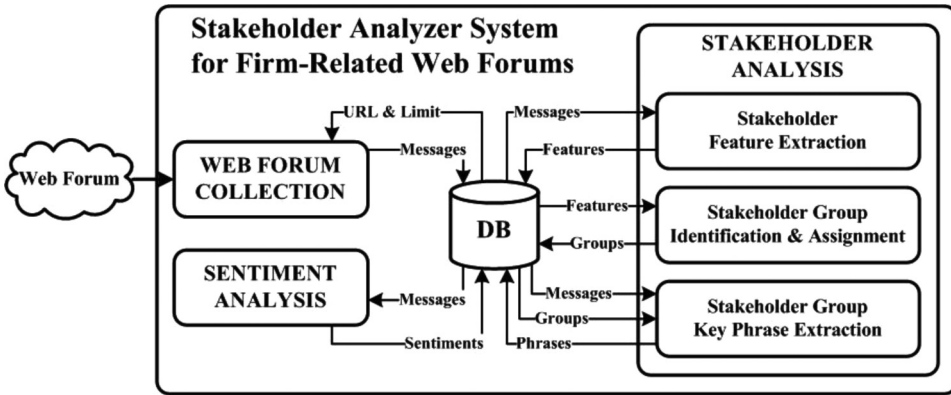


Fig. 1. Conceptual design of the SA system for firm-related web forums.

analysis does not presume generic functional role-based stakeholder groups but instead allows interest-based stakeholder groups with similar characteristics, interests, and perspectives on the firm to emerge from within the forum through unsupervised clustering. A stakeholder feature representation assessing the various functions of language and communication fulfilled by forum discussion and activity is defined to comprehensively evaluate the forum participants in the stakeholder analysis. Forum activity is accumulated over a time window to develop stable profiles for participants. To aid in the interpretation and understanding of the identified stakeholder groups, key phrases are extracted to characterize their frequent topics of discussion. To evaluate the opinions expressed by stakeholder groups in discussion, the SA also performs sentiment analysis on their messages.

Following design science, the SA was carefully designed based on the suggestions of the kernel theories and the stakeholder analysis problem characteristics. However, design is an inherently iterative search process for an effective solution to a problem [Hevner et al. 2004], and the SA does not represent the only solution to performing stakeholder analyses in firm-related web forums. We describe and justify the SA design and evaluate the SA in its intended application to demonstrate its effectiveness, but recognize that alternative designs exist for various aspects of the system. We encourage future research and design iterations to continue the design science process, building on our design or proposing alternatives, to advance this class of information systems for performing stakeholder analysis in firm-related web forums.

The SA is described in the subsequent sections. Following application of the SA to a forum, the contents of its database include information on messages and their sentiment scores, stakeholder features, groups, and membership assignments and key phrases characterizing their discussions. With these contents, the SA database can support various inquiries regarding the forum and the distinctive stakeholder groups participating in discussion. To evaluate the stakeholder analysis and information derived from the forums, the SA database is queried for daily measures at the forum and stakeholder group levels to apply in stock return prediction.

To account for the dynamic nature of forums and provide for near-real-time monitoring of stakeholder groups, the SA can be deployed over time using three approaches adapted from temporal text mining [Cao et al. 2006]. In the landmark approach, the SA is applied once to collect and analyze a forum reaching back in time to a specific landmark date. In the sliding window approach, the SA is applied iteratively at regular time intervals, each time incrementally updating the database with the most recently posted messages and performing the stakeholder analysis on only a limited

time window of history. The sliding window advances with time and older discussions are excluded, allowing recent participation in the forum to influence the stakeholder analysis. A sliding window approach has been similarly applied in temporal analyses of other social media, such as in Twitter trending topic detection [Pervin et al. 2013]. The fading window approach diminishes the influence of discussions over time by weighting their contributions to the stakeholder feature representation based on their age using a decay function. In this study, the SA is applied iteratively according to a sliding window approach to perform daily stakeholder analysis in each forum and stock return prediction. A landmark approach is also used to perform stakeholder analyses spanning the entire period. The fading window approach was not utilized, as it is suitable to rapidly changing forums or a longer time window. In both the sliding and landmark window approaches, the forum activities of participants are accumulated over the time window to develop stable profiles for the stakeholder analysis.

3.1. Firm-Related Web Forum Collection

The SA acquires data from forums of interest using a collection engine developed specifically for the system. The collection engine takes as input a target URL for the forum and a date and time in the past to limit the collection process. It then crawls the forum structure and collects all messages posted after the collection date and time and associated metadata. This feature allows the SA to first accumulate a large historical collection of the forum and then incrementally update its database to facilitate near-real-time monitoring by retrieving only the most recent messages posted since last collection. The collection engine consists of two main subcomponents, a forum spider and a message parser. These modules are tailored to the specific structure of the forum platform targeted for collection. The forum spider crawls through the forum structure acquiring web pages that contain the embedded messages. Web pages are then passed to the message parser, which extracts the messages and metadata and stores this information in the database, to be utilized by the analytic components of the SA.

3.2. Stakeholder Analysis

The stakeholder analysis component of the SA has three subcomponents, feature extraction, group identification and assignment, and key phrase extraction, described in the following sections. Prior to a stakeholder analysis, the forum activity of each participant is accumulated over the time window to develop a stable profile of their discussions and behavior. A stakeholder analysis in a forum is based on forum activity over the entire time window.

3.2.1. Stakeholder Feature Extraction. The SA utilizes three categories of features in a stakeholder analysis listed in Table I, content, social network, and writing-style-based features, corresponding to the ideational, interpersonal, and textual meta-functions of SFLT, respectively. These features form a comprehensive representation of forum participation to evaluate in the stakeholder analysis. Content and writing-style-based features are extracted for each participant in the forum based on the accumulation of his or her messages written during the time window of analysis. Social-network-based features are extracted based on all discussion threads joined by posting a message during the time window. To represent the ideational meta-function and the ideas expressed by stakeholders in forum discussion, word n -gram frequencies are extracted as content-based features. The unigram model is extended to uni-, bi-, and tri-grams to capture phrase-level content. Before extracting the word n -grams, words containing nonalphanumeric characters and simple function words are removed from the messages. Word (uni-, bi-, tri-) grams occurring in a minimum (1%, 0.1%, 0.05%) of messages in the time window are retained to remove infrequently occurring words and phrases

Table I. Stakeholder Feature Representation

SFLT Meta-Function	Feature Category	Subcategory	Features
Ideational	Content based	–	Word n -grams (uni-, bi-, tri-)
Interpersonal	Social network based	–	Interactions with participants in discussions
Textual	Writing style based	Lexical	Sentences per message Words per message Words per sentence Unique words per message Content words per message Unique content words per message Stop words per message Short words per message Word length frequency (20) Characters per message Characters per word Whitespace characters per message Uppercase characters per message Alphabetic character frequency (26) Special character frequency (20) Digit frequency (10) Hapax legomena Dis legomena Brunet W Honore H Sichel S
		Syntactic	POS n -grams (uni-, bi-, tri-) Punctuation frequency (8) Stop word frequency (448)

from the representation. The differing frequency thresholds were established to promote the inclusion of content-rich phrase-level grams. Word n -gram frequencies are scaled for each stakeholder to the message level to remove forum activity levels from the representation.

To represent the interpersonal meta-function and the forming of interpersonal connections between stakeholders, their interactions in forum discussion threads are extracted as social-network-based features. A social network and the ties between members are commonly represented using an interaction matrix. With n members in the forum social network, an n -by- n matrix is formed with each cell representing the presence, direction, or intensity of the relationship between two members. In a forum, relationships are formed through interaction in discussion. In this study, an interaction is defined as two participants posting a message to the same discussion thread within the time window of analysis. Repeated interactions with preferred participants over time may reveal homophily, similar interests and confirming perspectives, and mutually held stakes in the firm. To represent the intensifying of the interpersonal connections between stakeholders, each time two participants interact in a discussion thread, their cell value in the interaction matrix is incremented. The interaction vector for each participant is then scaled by his or her total number of interactions to remove forum activity levels from the representation.

To represent the textual meta-function and the expressions of style in the writing of stakeholders, writing-style-based features are extracted. These features may capture the de-individuation of stakeholders and conformance to their group norms, expressed through the adoption of similar writing styles and practices. The writing-style-based features are adopted from prior stylometric representations [Zheng et al. 2006; Abbasi and Chen 2008a, 2008b] and include lexical and syntactic measures of participant

writing. Lexical features include word-level, character-level, and vocabulary richness measures. Syntactic features include the frequency of punctuation and function words and part-of-speech grams extracted after applying the Stanford POS tagger [Toutanova et al. 2003]. The frequencies of the extracted style features are then scaled to the message, sentence, word, or character level, as described in Table I.

Although Table I lists many features corresponding to the textual meta-function, the stakeholder representation is designed with balanced emphasis on each SFLT meta-function and to allocate approximately equal feature space to each meta-function in a stakeholder analysis. A wide variety of writing-style-based features are adopted from prior stylometric representations [Zheng et al. 2006; Abbasi and Chen 2008a, 2008b] to counterbalance the numerous word n -grams and participants in the forum social networks that represent the ideational and interpersonal meta-functions. To evaluate the balance in representation across meta-functions, the numbers of features applied in the stakeholder analyses are presented with the results.

3.2.2. Stakeholder Group Identification and Assignment. The SA utilizes the extracted stakeholder features to perform a stakeholder analysis on participants in a forum. Researchers have raised concerns that functional role-based stakeholder models are overly simplistic [Gioia 1999] and instead have favored unsupervised clustering for stakeholder group identification and assignment [Wolfe and Putler 2002]. Following the stakeholder analysis literature, the SA utilizes an unsupervised clustering approach to allow distinctive interest-based stakeholder groups to emerge from the forum based on similarities in participants captured in the comprehensive stakeholder feature representation. An unsupervised approach requires no prior knowledge of the firm, automates the challenging stakeholder group identification step, and allows for the discovery of previously unknown stakeholder groups.

Although many unsupervised clustering approaches are available from the machine-learning literature, two insights from the stakeholder theory guide the selection of the approach utilized by the SA. Stakeholders may belong to more than one stakeholder group [Dill 1975], and the number of distinctive interest-based stakeholder groups participating within a forum is unknown in advance. A clustering approach supporting these characteristics is probabilistic clustering using a Gaussian mixture model and the expectation maximization algorithm for parameter estimation [Witten and Frank 2005]. The SA utilizes the Weka data-mining package implementation of this clustering algorithm [Witten and Frank 2005]. Probabilistic approaches model instances with probabilities of assignment to each cluster without making exclusive cluster divisions. Using a Gaussian mixture model, a near-optimal number of clusters may be determined without prior specification using maximum likelihood estimation [Smyth 2000]. This approach is adopted by the SA to identify the number of distinctive stakeholder groups in a forum based on the comprehensive stakeholder feature representation. After clustering, the probabilistic assignments of forum participants to each of the identified stakeholder groups (continuous values from 0 to 1, with a sum of 1) are stored in the SA database to be utilized in the key phrase extraction and in generating stakeholder-group-level measures of a forum.

Probabilistic Gaussian mixture models, like many other clustering approaches, assume the independence of features in the input representation [Witten and Frank 2005]. However, features in the comprehensive stakeholder representation may be correlated or interdependent. Therefore, principal components are extracted from the stakeholder feature representation prior to clustering. The SA also utilizes the Weka data-mining package implementation of principal component analysis [Witten and Frank 2005]. Principal component analysis (PCA) transforms the stakeholder representation feature space based on the variance observed across participants, increasing

the independence of input features and reducing the dimensionality of the final representation used for clustering. Clustering approaches applied to document collections often incorporate similar feature reduction techniques to cope with the high dimensionality of text-based representations. In PCA, principal components are computed using 10 features per component, and the top components accounting for at least 90% of the variance observed across participants are retained to form the final representation used for stakeholder clustering.

3.2.3. Stakeholder Group Key Phrase Extraction. To interpret the results of a stakeholder analysis and aid in the understanding of the stakeholder groups identified within a forum, the SA extracts key phrases to characterize their frequent topics of discussion. The key phrase extraction component utilizes the content-based features for each participant and his or her stakeholder group probabilities assigned by the clustering. Multiple stakeholder groups may be characterized by similar key phrases. The key phrases are simply extracted by the SA to determine the frequent topics of discussion of the identified stakeholder groups and to better understand and describe their participation in a forum. The key phrase extraction does not account for distinctive social-network-based or writing-style-based features, which may be responsible for the division of stakeholder groups in the clustering, despite having similar topics of discussion. The stakeholder clustering also produces overlapping groups and probabilistic assignments, and multiple groups may discuss the same topics and issues in the forum from different perspectives. In future research and design iterations of the SA, the key phrase extraction component will be extended to also extract the distinctive social-network-based and writing-style-based features characterizing each group.

To determine the key phrases representative of stakeholder group discussion, their weighted frequencies are calculated and ranked. The word n -gram frequencies for each participant are scaled by the probabilities of their assignment to each stakeholder group and accumulated across all participants in the forum to generate the keyword values, as defined later. Word n -grams with high keyword values for a particular stakeholder group are frequently used by its members in forum discussion, potentially indicating their interests and stakes in the firm. Once calculated, the keyword values are stored in the SA database and can be retrieved to characterize the frequent topics of discussion of a stakeholder group.

$$K_{ws} = \sum^P (F_{wp} * M_{ps}),$$

where:

K_{ws} = keyword value for word n -gram w and stakeholder group s

Firm-related web forum participant p from $(1 \dots P)$

F_{wp} = frequency of usage of word n -gram w by forum participant p

M_{ps} = probability of forum participant p 's assignment to stakeholder group s

3.3. Stakeholder Sentiment Analysis

To evaluate the opinions expressed by stakeholder groups in forum discussion, the SA performs sentiment analysis on their messages. Instead of adopting a restrictive sentiment classification scheme designed to distinguish investment positions as in prior studies [Antweiler and Frank 2004; Das and Chen 2007], the SA utilizes a general approach intended to provide the flexibility to evaluate various stakeholder groups and their diverse expressions of sentiment on the firm. We describe the SA sentiment analysis component but recognize that many design alternatives may exist from the extensive sentiment analysis literature. To demonstrate the effectiveness of the SA, we evaluate its performance in forum message sentiment classification and present these results in Section 4.1. In future research and design iterations of the SA, alternative

approaches to stakeholder sentiment analysis will be devised and examined. The SA utilizes the general-purpose Opinion Finder (OF) lexicon to measure the sentiment and subjectivity expressed in each forum message [Wilson et al. 2005]. The OF lexicon consists of 7,630 words carefully culled by linguists and their respective sentiment and subjectivity classifications. Lexicons created for application on professionally written financial statements and disclosures issued by the firm, such as 10-Ks [Loughran and McDonald 2011], are less suited to evaluate the casual and diverse discussions occurring in the forums. The OF lexicon has been successfully applied in other social media studies, including in Twitter sentiment analysis to indicate stock market movements [Bollen et al. 2011], and in Amazon.com product review spam detection [Lau et al. 2011]. Each word in the OF lexicon is classified as expressing either strongly positive, weakly positive, strongly negative, or weakly negative sentiments (opinions) and either strong or weak subjectivity. The lexicon also specifies the part of speech for the words in their intended usage. To ensure the OF lexicon is correctly applied, the SA performs part-of-speech tagging using the Stanford POS tagger [Toutanova et al. 2003]. It then scans through the tagged message text for occurrences of words in the OF lexicon applied in their correct part of speech and accumulates sentiment and subjectivity scores, as defined later. Sentiment is the count of the number of positive words minus the number of negative words used in the message. Subjectivity is the number of subjective words used in the message. Other researchers have used similar measures of sentiment when applying the OF lexicon, where the occurrence of a positive word increments a sentiment score and a negative word decrements the score [Bollen et al. 2011]. The sentiment and subjectivity measures were not normalized to allow longer messages expressing strong sentiments to have a greater impact. Term frequency is commonly utilized in sentiment analysis feature representations [Pang and Lee 2008], including those applied in web forums [Abbasi et al. 2008c]. While the authors of the OF lexicon have provided intensity (strong/weak) classifications for each word, researchers have failed to incorporate this information in their applications of the lexicon, treating strong and weak sentiment words equivalently [Bollen et al. 2011]. We favor utilizing these intensity classifications in our application of the OF lexicon. Therefore, we assign words classified in the OF lexicon as strongly positive, negative, or subjective twice the value of words classified as weakly positive, negative, or subjective in the sentiment and subjectivity measures. After sentiment analysis, the sentiment and subjectivity scores for each message are stored in the SA database and utilized to generate sentiment measures for the stakeholder groups in a forum.

$$\begin{aligned} sentiment_m &= (2SP_m + WP_m) - (2SN_m + WN_m) \\ subjectivity_m &= 2SS_m + WS_m, \end{aligned}$$

where:

- $sentiment_m$ = sentiment expressed in message m
- $subjectivity_m$ = subjectivity expressed in message m
- SP_m = count of strongly positive OF words used in message m
- WP_m = count of weakly positive OF words used in message m
- SN_m = count of strongly negative OF words used in message m
- WN_m = count of weakly negative OF words used in message m
- SS_m = count of strongly subjective OF words used in message m
- WS_m = count of weakly subjective OF words used in message m

4. EXPERIMENTS AND RESEARCH TESTBEDS

To examine whether distinctive stakeholder groups with shared characteristics, interests, and perspectives on the firm can be identified within firm-related web forums, we performed stakeholder analyses on six firms and forums using the SA over a 3-month

Table II. Classifications of Domain Expert for Sentiment Analysis Evaluation

	Positive	Negative	Neutral	Negation	Literary Device
AMZN	23%	32%	45%	23%	26%
BA	16%	44%	40%	26%	32%
F	15%	47%	38%	30%	37%
HPQ	20%	47%	33%	35%	22%
INTC	24%	45%	31%	33%	30%
WMT	16%	42%	42%	18%	29%
Overall	19%	42.8%	38.2%	27.5%	29.3%

Table III. Performance of SA in Forum Message Sentiment Classification

	Precision	Recall	F-Measure	Negation Acc.	Literary Devices Acc.
AMZN	0.80	0.75	0.78	70%	54%
BA	0.80	0.73	0.76	65%	59%
F	0.81	0.70	0.75	63%	62%
HPQ	0.83	0.78	0.81	71%	73%
INTC	0.85	0.82	0.84	76%	73%
WMT	0.77	0.73	0.75	61%	52%
Overall	0.81	0.75	0.78	68%	62%

period from October 30, 2012, to January 31, 2013. Following prior studies, forums hosted on Yahoo Finance were selected [Antweiler and Frank 2004; Das and Chen 2007]. The six firms are major companies representing a cross-section of industries, with stocks heavily traded in the stock market, and some of the most popular forums on Yahoo Finance: Amazon (AMZN), Boeing (BA), Ford (F), Hewlett-Packard (HPQ), Intel (INTC), and Wal-Mart (WMT). We focused on these six major firms to demonstrate stakeholder analysis over a diverse and representative sample but recognize that the portfolio and time period of analysis is limited, and the findings of this study may not generalize to all other firms or time periods.

4.1. Sentiment Analysis Evaluation

To first demonstrate the effectiveness of the SA sentiment analysis component, we evaluate its performance in the sentiment classification of forum messages. One hundred messages from each of the six firm-related forums were randomly selected from all messages posted during the period of analysis and collected for the sentiment classification experiments. The 600 forum messages were carefully evaluated by a domain expert and classified into one of three sentiment classes (positive/negative/neutral). Since many complex aspects of writing known to affect sentiment analysis performance are not addressed by the SA approach, the domain expert also noted the use of negation and literary devices (including sarcasm, metaphors, and analogies) to evaluate the SA performance when these complex aspects were used in the messages. The classifications determined by the domain expert for the 100 messages sampled from each forum are summarized in Table II. Forum messages were evaluated as mostly negative, and the usage of both negation and complex literary devices was common, present in about a third of messages.

The SA performed sentiment analysis on the 600 forum messages, and classifications were determined using the sentiment scores. Messages assigned a sentiment score greater than zero were classified positive, less than zero were negative, and zero were neutral. The results of the experimentation in sentiment classification are presented in Table III, including precision, recall, and F-measure statistics across all sentiment

classes and classification accuracies in the cases where negation or complex literary devices were used in the forum messages. Across the six forums, the SA sentiment analysis approach performed reasonably well, accurately classifying 75% of messages and demonstrating fairly good precision across all sentiment classes. Classification accuracy degraded to 68% and 62% when negation or complex literary devices were used in the forum messages, respectively. In future design iterations, the SA sentiment analysis approach will be improved to better account for these aspects of writing. Sentiment class-level precision, recall, and F-measure statistics are reserved for Appendix A. These results show that the SA was relatively weak in positive class precision and negative class recall, possibly due in part to the thresholds in sentiment score selected for the positive, negative, and neutral class boundaries. On the whole, these experiments demonstrate that the SA performs reasonably well in sentiment analysis and is capable of evaluating stakeholder sentiment expressed in discussion.

4.2. Stock Return Prediction

To evaluate the information conveyed within the firm-related web forums and the additional refinement provided by stakeholder analysis, we followed prior research [Antweiler and Frank 2004; Das and Chen 2007] and examined the relationships between measures extracted from the forum discussions and firm stock return on the subsequent trading day, and also applied them in the prediction of daily stock return. In each form of evaluation, we establish baseline models following the technician approach in the stock prediction literature consisting of established predictors of firm stock return, including the prior trading day's stock return, volatility, and trading volume. Other researchers have used prior stock values or return [Barnes et al. 2000; Pai and Lin 2005] and volatility [Pesaran and Timmerman 1995; Hadavandi et al. 2010] in similar models. To determine the incremental benefit of including information derived through the analysis of the forums, forum and stakeholder models are also developed that supplement the baseline variables with measures extracted by the SA. The forum models include measures at the forum level, without considering the stakeholder analysis. These models follow prior studies that have assumed a homogenous forum population [Antweiler and Frank 2004; Das and Chen 2007]. Alternatively, the stakeholder models include measures extracted at the stakeholder group level, to incorporate their distinctive information on the firm. The refined assessment of the forums provided by the stakeholder analyses should produce measures that better explain and predict subsequent firm stock returns than aggregate forum-level information.

The overall period of analysis spanned 3 months and 62 trading days, from October 30, 2012, to January 31, 2013. This exceeds the lengths of similar studies that have focused on a 1- or 2-month time period [Das and Chen 2007; Schumaker and Chen 2009a; Schumaker et al. 2012]. Stock return prediction and simulated trading of the six firm stocks were performed for 31 consecutive trading days, from December 17, 2012, to January 31, 2013. The SA was deployed following market close on December 16, 2012, at 4:00 p.m. EST according to a sliding window approach, with 31 trading days of history (reaching back to October 30) included in the forum analyses and stock return modeling, to generate predictions for the 17th. Throughout the stock return prediction period, before each subsequent trading day the SA was deployed to incrementally update the forum collections, advancing the sliding window one trading day, and performing the forum analyses and stock return modeling again using only the most recent 31 trading days of history. Following the completion of stock return prediction on January 31, 2013, the SA was also applied to analyze each forum reaching back to the landmark date of October 30, to conduct regression analyses and examine the relationships between measures extracted from the forums and subsequent firm stock returns over the 62-trading-day period.

Table IV. Stock and Stakeholder Analyzer System Variables

Source	Variable	Abbreviation	Description
Stock	Return	RET	Daily stock return
	MKT Return	MKT	Daily market return (DJI)
	Volatility	VOLA	Daily stock volatility
	Volume	VOLU	Daily stock trading volume (in log)
Stakeholder Analyzer System	Num. Messages	MESS	Number of messages posted on a day (in log)
	Avg. Length	AVG LEN	Average length of messages posted on a day
	Std. Dev. Length	DEV LEN	Std. dev. in length of messages posted on a day
	Avg. Subjectivity	AVG SUBJ	Average subjectivity in messages posted on a day
Forum or Stakeholder Group	Std. Dev. Subjectivity	DEV SUBJ	Std. dev. in subjectivity in messages posted on a day
	Avg. Sentiment	AVG SENT	Average sentiment in messages posted on a day
	Std. Dev. Sentiment	DEV SENT	Std. dev. in sentiment in messages posted on a day

4.2.1. Description of Stock and Stakeholder Analyzer System Variables. The stock and SA system variables used in the regression analyses and stock return prediction models are shown in Table IV. Abbreviations for the variables are used in the model specification in Section 4.2.2, and complete variable names are presented in the regression results.

The daily stock data used in the research was also retrieved from Yahoo Finance. Daily stock return is defined as the difference in share price from the prior day's market close to today's market close divided by the prior day's market close. Market return is the daily return of the Dow Jones Industrial index. The daily volatility of a stock is defined as the difference in the high and low of a stock price, divided by the average of the open and closing prices, as in Das and Chen [2007]. The daily trading volume is the number of shares traded that day, in log form.

Following the stakeholder analysis of each firm-related forum in each time window, the SA database was queried for daily measures at the forum and stakeholder group levels, which were applied to explain and predict subsequent firm stock return. The seven daily measures included the number of messages posted and the average and standard deviation in message length, subjectivity, and sentiment expressed on the day preceding a trading day. Prior studies have evaluated similar forum measures, including message volume and length, and the average and disagreement in sentiment [Antweiler and Frank 2004; Das and Chen 2007]. Message length is measured by the total number of words. Message subjectivity and sentiment are defined in the SA description. To coincide with the closing of the market and align with the stock variables, daily measures are defined using trading days that begin and end at 4 p.m. EST.

4.2.2. Stock Return Regression Analysis. The relationships between the prior stock and SA variables and subsequent daily firm stock return were evaluated using multiple linear regression models, following prior studies [Antweiler and Frank 2004; Das and Chen 2007]. The baseline, forum, and stakeholder stock return models are defined later and developed for each firm. The dependent variable in each model is firm stock return on trading day t . The baseline model consists of five established indicators of stock return, including the prior market return and prior firm stock return, volatility, and trading volume. A concurrent market return variable is also included to capture variance in the firm stock return associated with the market. The forum model adds the seven daily forum measures to the baseline stock variables. Alternatively, the stakeholder model supplements the baseline stock variables with the same seven measures as in the forum model but extracted by the SA at the stakeholder group level (resulting in seven measures per group). For the stock variables, the prior day (day $t-1$) refers to

the prior trading day. For the SA variables, day $t-1$ refers to the prior calendar day, to utilize the most recent forum discussions.

Performing stakeholder analysis in a forum may reveal groups with more relevant information on firm stock return than others. And stakeholder groups and measures with less relevance in explaining stock return should be removed from the models. Therefore, stakeholder models are developed by first selecting from the available stakeholder measures extracted by the SA, retaining only the most relevant groups and indicators of stock return. Specifically, stepwise regressions and model selection are performed, searching among the SA stakeholder group level variables bidirectionally and evaluating candidate models using the Akaike information criterion (AIC) [Akaike 1973]. Variables are selected from the stakeholder group measures only, whereas all stock variables are retained for direct comparison with baseline and forum models.

Baseline Models

$$RET_t = \beta_0 + \beta_1 MKT_t + \beta_2 MKT_{t-1} + \beta_3 RET_{t-1} + \beta_4 VOLA_{t-1} + \beta_5 VOLU_{t-1} + \varepsilon_t$$

Forum Models

$$RET_t = \beta_0 + \beta_1 MKT_t + \beta_2 MKT_{t-1} + \beta_3 RET_{t-1} + \beta_4 VOLA_{t-1} + \beta_5 VOLU_{t-1} \\ + \beta_6 MESS_{t-1} + \beta_7 AVG LEN_{t-1} + \beta_8 DEV LEN_{t-1} + \beta_9 AVG SUBJ_{t-1} \\ + \beta_{10} DEV SUBJ_{t-1} + \beta_{11} AVG SENT_{t-1} + \beta_{12} DEV SENT_{t-1} + \varepsilon_t$$

Stakeholder Models

$$RET_t = \beta_0 + \beta_1 MKT_t + \beta_2 MKT_{t-1} + \beta_3 RET_{t-1} + \beta_4 VOLA_{t-1} + \beta_5 VOLU_{t-1} \\ + \beta_6 MESS_{1t-1} + \beta_7 AVG LEN_{1t-1} + \beta_8 DEV LEN_{1t-1} + \beta_9 AVG SUBJ_{1t-1} \\ + \beta_{10} DEV SUBJ_{1t-1} + \beta_{11} AVG SENT_{1t-1} + \beta_{12} DEV SENT_{1t-1} + \dots \\ + \beta MESS_{St-1} + \beta AVG LEN_{St-1} + \beta DEV LEN_{St-1} + \beta AVG SUBJ_{St-1} \\ + \beta DEV SUBJ_{St-1} + \beta AVG SENT_{St-1} + \beta DEV SENT_{St-1} + \varepsilon_t$$

4.2.3. Stock Return Prediction and Simulated Trading. The stakeholder analyses performed by the SA were also evaluated by applying the measures extracted from the forums in the true prediction of daily firm stock return. To directly link the stock return prediction to the regression analyses, the same models and approaches were utilized. Multiple linear regression models were developed by applying the same variables as specified in the baseline, forum, and stakeholder regression models in Section 4.2.2, although the concurrent market return variable was removed to perform true prediction of future firm stock return using only past information. Stepwise regressions and model selection based on AIC were performed, as in the regression analysis. However, in the stock return prediction, all models utilized model selection, and variables were selected from among all stock and SA variables. While the regression analyses spanned the entire 62-trading-day period, to perform stock return prediction over time, the SA utilized the sliding window approach described in Sections 3 and 4.2, including a 31-trading-day time window in the stakeholder analyses and stock return modeling.

Three criteria evaluate model performance in predicting stock return, the closeness of the prediction, directional accuracy, and simulated trading of firm stock [Schumaker and Chen 2009a, 2009b]. The closeness of the prediction to the actual stock return on the trading day is measured using mean absolute error (MAE). Directional accuracy evaluates performance in predicting the direction of the stock return, positive or negative. Of the two, directional accuracy is considered more important since trading strategies are often dependent upon the direction of the predictions. In the simulated trading, firm stock is traded based on the daily predictions of the models, to determine the economic value of applying the information extracted by the SA from the forums. If the models predicted positive (negative) returns for the upcoming trading day, firm stock was bought (shorted) and then sold at the end of the day.

Table V. Descriptive Statistics for Firm Stocks and Forum Collections

Descriptive Statistics		AMZN	BA	F	HPQ	INTC	WMT
Stock	Average Daily Return	0.0022	0.0009	0.0026	0.0033	-0.0001	-0.0009
	Average Daily Volatility	0.0225	0.0163	0.0233	0.0334	0.0185	0.0126
	Average Daily Trading Volume	6.49	6.73	7.69	7.44	7.67	6.91
Forum	Total Messages	10,804	8,596	19,497	7,289	7,482	9,782
	Total Participants	1,524	933	1,948	1,169	1,313	902
	Average Daily Messages	174	138	314	117	120	157
	Average Message Length	44	69	45	52	71	76
	Average Message Subjectivity	2.33	3.95	2.54	2.86	3.22	3.74
	Average Message Sentiment	0.17	-0.08	-0.15	0.00	0.52	0.18

Table VI. Number of Features Included in the Stakeholder Representations and Principal Components Retained in the Principal Component Analysis

SFLT Meta-Function	Ideational	Interpersonal	Textual	Total	Principal Components
Feature Category	Content Based	Social Network Based	Writing Style Based		
AMZN	921	1,524	699	3,144	629
BA	1,481	933	835	3,249	459
F	863	1,948	717	3,528	765
HPQ	1,072	1,169	741	2,982	503
INTC	2,895	1,313	819	5,027	516
WMT	2,542	902	840	4,284	327

5. RESULTS

The SA collected all discussions from the six firm-related forums from October 30, 2012, to January 31, 2013, including 63,450 messages posted by 6,372 unique participants on the days preceding the 62 trading days included in the stock prediction and regression analysis. Statistics describing the firm stocks and six 62-day forum collections are presented in Table V. Discussion in each forum was highly active, with more than 100 messages posted per day. The Ford forum collection was particularly large, with over 19,000 messages from 1,900 participants. Messages in the Wal-Mart and Intel forums tended to be longer, averaging 76 and 71 words per message, respectively, compared to the shorter messages in the Amazon and Ford forums (44 and 45, respectively). Messages in the Boeing forum expressed the highest subjectivity (3.95). Discussion in the Intel forum was the most positive with an average message sentiment score of 0.52, while negativity was commonly expressed in the Ford forum (-0.15).

5.1. Stakeholder Analyses

Before presenting the results of the stock return regression analyses and prediction, we describe the results of the stakeholder analyses performed in each of the six forums over the entire 62-trading-day period. The numbers of features included in the stakeholder representations for each firm are presented in Table VI. Although a wide variety of writing-style-based features were incorporated into the stakeholder representation, they were fairly comparable in number to the content-based and social-network-based features for each firm. Based on the observed numbers representing each feature category, the design goal of balanced emphasis on each SFLT meta-function in the stakeholder representation was generally achieved. Principal components were extracted from the stakeholder feature representations for each forum to reduce the dimensionality and increase the independence of input features prior to stakeholder clustering. The PCA retained the number of principal components required to account for at least

Table VII. Stakeholder Group Clustering Results

	AMZN	BA	F	HPQ	INTC	WMT
Stakeholder Group Clusters	6	5	6	5	5	3
Log-Likelihood Stakeholder Group Clusters	-1,027.2***	-868.8***	-1,205.4***	-798.9***	-848.7***	-587.9***
Log-Likelihood Single Cluster	-1,173.2	-971.5	-1,391.5	-988.9	-1,118.6	-759.8

Likelihood ratio test: improvement over single cluster model at ***p < 0.001.

Table VIII. Wal-Mart Stakeholder Group Key Phrase Extraction Results

Wal-Mart	Avg. Prob.	Keywords	Key Phrases
Stakeholder Group 1 "information exchange"	0.50	good thanks great look news sales	stock market made money great job post topic press release good news
Stakeholder Group 2 "stores and expansion"	0.13	open stores workers state store union	stores open private sector china india walmart stores unemployment benefits nation largest
Stakeholder Group 3 "labor and union"	0.34	union workers pay money work days	food stamp union workers minimum wage work thanksgiving health care pay workers

90% of the variance in the feature space observed across participants, also listed for each firm in Table VI. The unsupervised clustering of stakeholders was performed for each firm using the principal components. The numbers of distinctive interest-based stakeholder groups identified within each forum through the clustering are presented in Table VII, ranging from three to six for each firm. Very small stakeholder groups with less than 5% of participants were removed by the SA. The log-likelihood statistics for the stakeholder group clustering models are also reported in Table VII for each firm. We compare these statistics to the log-likelihoods of models that represent participants with a single cluster or stakeholder group, as in prior studies. Likelihood ratios tests were conducted, and the stakeholder group clustering models were found to provide statistically significant improvements in representing the distribution of participants within the feature space.

To aid in the understanding of the identified interest-based stakeholder groups, the SA extracted key phrases to characterize the frequent topics of discussion of each group. The keywords and phrases extracted for the identified stakeholder groups of Wal-Mart are presented in Table VIII, with the average probability of each group's representation in the forum. Similar tables for the other firms and stakeholder analyses are presented in Appendix B. Stakeholder groups may be characterized by the same key phrases, since their divisions may be based on differences in social-network-based or writing-style-based features, or they may discuss the same topics and issues from different perspectives. Future design iterations will extend the key phrase extraction component of the SA to extract distinctive social-network-based and writing-style-based features characterizing each group. The average probability of a stakeholder group is defined as the sum of all probabilistic assignments to that group across forum participants,

divided by the total number of participants. Each keyword and phrase listed was ranked in the top 10 in keyword value for its respective stakeholder group. Based on scrutiny of the keywords and phrases, each stakeholder group was assigned a descriptive label characterizing its frequent topics of discussion. These stakeholder group labels are also included in the presentation of the stakeholder regression models to improve the interpretability of the results.

In the Wal-Mart forum, the largest interest-based stakeholder group had an average probability of 0.50 and, as shown by their keywords, often engaged in discussion to exchange information, news, and press releases on the firm. A second stakeholder group (0.13 average probability) often discussed the expansion of the firm and opening of new Wal-Mart stores. Finally, a third stakeholder group (0.34) focused its discussion on labor- and union-related issues. Based on scrutiny of the extracted key phrases, these interest-based stakeholder groups of Wal-Mart were assigned the labels “information exchange,” “stores and expansion,” and “labor and union,” characterizing their forum discussion and expressed interests in the firm.

The stakeholder analysis and key phrase extraction identified diverse groups in the Wal-Mart forum expressing their distinctive interests in the firm, issues relevant to the performance of the firm. Distinctive stakeholder groups were also identified in the other forums, reflected in the key phrases characterizing their discussion presented in Appendix B. Descriptive labels were similarly assigned to the stakeholder groups identified in the other forums through scrutiny of the extracted key phrases. The key phrases also revealed that discussion among participants in the six forums was general and diverse, and usage of the specialized terminology and language in finance sentiment lexicons [Das and Chen 2007; Loughran and McDonald 2011] was limited.

5.2. Stock Return Regression Analysis

The relationships between measures extracted by the SA from the forum discussions and firm stock return on the next trading day were examined through regression analyses spanning the 62 trading days. Prior to the regression analyses, the correlations of the stock and SA variables were examined. The results of the correlation analyses are reserved for Appendix C. At the forum level, the seven SA variables were mildly correlated with subsequent firm stock return, statistically insignificant in most cases, but correlations at 10% or greater were observed for each firm. Interestingly, nearly all measures of the Ford forum were significantly correlated with stock return, at levels comparable to the concurrent market return.

Following the technician approach to stock return prediction, baseline models were developed including several known predictors of stock return, as defined in Section 4.2.2. The results of the baseline regression analyses are reserved for Appendix D. Expectedly, the majority of variance in firm stock returns was explained by the concurrent market return. The portfolio consists of large capitalization stocks that generally follow the movements of the market. Prior market return was also statistically significant in the Amazon model, and prior volatility was significant in the Intel and Wal-Mart models. The adjusted R^2 values for the baseline regression models ranged from 0.144 (Ford) to 0.336 (Intel).

To compare with the baseline models and determine the incremental benefit of including information extracted from the forums, forum models were also developed as defined in Section 4.2.2. The forum models add measures of discussion at the forum level of analysis, without considering the stakeholder groups. The results of the forum regression models are presented in Table IX for each firm. The variable coefficients and t-values are listed for each term in the regressions. In two forum models (Boeing and Hewlett-Packard), forum-level measures provided significant indications of subsequent firm stock return. Information from the Hewlett-Packard forum was particularly

Table IX. Forum Stock Return Regression Analysis Results

Forum Stock Return Regression		AMZN	BA	F	HPQ	INTC	WMT
		Return _t	Return _t	Return _t	Return _t	Return _t	Return _t
Stock	Intercept	−0.003 (−0.021)	−0.012 (−0.217)	−0.094 (−0.843)	0.099 (0.534)	0.093 (1.032)	0.023 (0.367)
	MKT Return _t	1.021*** (3.123)	0.993**** (4.926)	0.839*** (2.838)	1.672**** (3.990)	1.047**** (4.231)	0.671**** (4.027)
	MKT Return _{t−1}	0.663* (1.989)	−0.107 (−0.438)	0.210 (0.715)	−0.390 (−0.829)	−0.284 (−1.007)	0.131 (0.699)
	Return _{t−1}	−0.162 (−1.182)	−0.000 (−0.002)	0.002 (0.020)	−0.222* (−1.799)	−0.012 (−0.092)	−0.030 (−0.200)
	Volatility _{t−1}	−0.122 (−0.320)	0.251 (1.026)	−0.121 (−0.417)	0.422 (1.241)	−0.720** (−2.576)	−0.256 (−0.885)
	Volume _{t−1}	0.003 (0.149)	0.002 (0.266)	0.016 (0.911)	−0.009 (−0.352)	−0.017 (−1.367)	−0.000 (−0.029)
Forum	Num. Messages _{t−1}	−0.002 (−0.650)	0.000 (0.549)	−0.006 (−0.789)	−0.007 (−1.469)	0.007 (1.591)	−0.001 (−0.480)
	Avg. Length _{t−1}	−0.000 (−1.494)	−0.000 (−1.178)	0.000 (0.061)	0.002*** (2.864)	−0.000 (−0.741)	0.000 (0.348)
	Std. Dev. Length _{t−1}	0.000 (0.723)	0.000 (1.665)	0.000 (0.287)	−0.000*** (−3.289)	0.000 (1.219)	0.000 (0.541)
	Avg. Subjectivity _{t−1}	0.012 (1.321)	0.002 (0.612)	0.002 (0.122)	−0.042*** (−3.353)	0.011 (1.468)	−0.007 (−1.289)
	Std. Dev. Subjectivity _{t−1}	−0.000 (−0.549)	−0.000* (−1.701)	−0.000 (−0.775)	0.002*** (3.267)	−0.000 (−1.438)	0.000 (1.041)
	Avg. Sentiment _{t−1}	0.002 (0.376)	0.003 (1.051)	−0.006 (−0.507)	−0.003 (−0.457)	0.002 (0.416)	−0.003 (−0.811)
	Std. Dev. Sentiment _{t−1}	−0.001 (−1.318)	0.000 (0.081)	0.000 (0.643)	0.001* (1.707)	0.000 (0.026)	−0.000 (−0.633)
R ²		0.299	0.404	0.339	0.449	0.443	0.389
Adjusted R ²		0.128	0.258	0.177	0.315	0.307	0.239

*p < 0.1; **p < 0.05; ***p < 0.01; ****p < 0.001.

powerful; both message length and subjectivity measures and the deviation in sentiment preceding a trading day were found statistically significant, and the adjusted R² value improved from 0.213 in the baseline model to 0.315 in the forum model. Overall, the inclusion of measures extracted at the forum level of analysis provided little benefit in explaining subsequent stock return, and the adjusted R² values for the forum regression models did not improve substantially, ranging from 0.128 (Amazon) to 0.315 (Hewlett-Packard).

To compare with the baseline and forum models and determine the additional refinement provided by the stakeholder analyses, stakeholder models were also developed as defined in Section 4.2.2. The stakeholder models replaced forum-level measures with stakeholder group level measures extracted by the SA to incorporate their distinctive information on the firm in the prediction of stock return. The refined assessment of the forums provided by the stakeholder analyses should produce measures that better explain and predict subsequent firm stock returns than aggregate forum-level information. Stakeholder models were developed for each of the six firms through stepwise regression and model selection using AIC. The results of the final stakeholder regression models are presented in Appendix E. As an example, the Wal-Mart stakeholder model is presented in Table X. In the Wal-Mart forum model, none of the forum-level variables were identified as significantly related to subsequent firm stock return. However, numerous stakeholder-group-level variables extracted by the SA were highly significant in the stakeholder regression model. Similarly impressive results were

Table X. Wal-Mart Stakeholder Stock Return Regression Analysis Results

Stakeholder Stock Return Regression		WMT Return _t
Stock	Intercept	0.010 (0.184)
	MKT Return _t	0.880 (5.084)****
	MKT Return _{t-1}	0.012 (0.069)
	Return _{t-1}	0.062 (0.426)
	Volatility _{t-1}	0.051 (0.184)
	Volume _{t-1}	0.001 (0.134)
Stakeholder Group 1 “information exchange” Avg. Prob.: 0.50	Num. Messages _{t-1}	0.006 (2.767)***
	Std. Dev. Subjectivity _{t-1}	0.001 (1.728)*
	Std. Dev. Sentiment _{t-1}	-0.002 (-2.419)**
Stakeholder Group 2 “stores and expansion” Avg. Prob.: 0.13	Avg. Length _{t-1}	-0.000 (-2.722)***
	Std. Dev. Length _{t-1}	0.000 (1.856)*
	Avg. Sentiment _{t-1}	-0.005 (-1.708)*
	Std. Dev. Sentiment _{t-1}	-0.000 (-1.474)
Stakeholder Group 3 “labor and union” Avg. Prob.: 0.34	Num. Messages _{t-1}	-0.005 (-2.770)***
	Std. Dev. Length _{t-1}	-0.000 (-2.451)**
	Avg. Subjectivity _{t-1}	-0.001 (-1.377)
	Std. Dev. Subjectivity _{t-1}	0.000 (3.624)****
	Avg. Sentiment _{t-1}	-0.002 (-2.343)**
	Std. Dev. Sentiment _{t-1}	0.000 (2.424)**
R ²		0.575
Adjusted R ²		0.397

*p < 0.1; **p < 0.05; ***p < 0.01; ****p < 0.001.

Table XI. Summary of Adjusted R² Statistics for Stock Return Regression Models

Adjusted R2	AMZN	BA	F	HPQ	INTC	WMT	Average
Baseline	0.164	0.240	0.144	0.213	0.336	0.179	0.213
Forum	0.128	0.258	0.177	0.315	0.307	0.239	0.237
Stakeholder	0.826	0.391	0.795	0.614	0.531	0.397	0.592

observed in the stakeholder models of the other firms, presented in Appendix E. In the Wal-Mart stakeholder model, measures of the group frequently discussing “labor and union” topics were highly indicative of subsequent firm stock return, with five group measures identified as statistically significant. As shown in the stakeholder model results for each firm, performing a stakeholder analysis refined the assessment of each forum, yielding distinctive stakeholder groups and measures that better explain subsequent firm stock returns. Stakeholder regression-model-adjusted R² values increased for each firm, ranging from 0.391 (Boeing) to an impressive 0.826 (Amazon).

A summary of the adjusted R² values for the baseline, forum, and stakeholder stock return regression models are presented in Table XI. Overall, the baseline and forum models explained relatively little of the variance in firm stock returns, although the low adjusted R² values are consistent with prior studies. The similar forum-level predictive model in Das and Chen [2007] produced an R² value of 0.07, and the contemporaneous stock return model in Antweiler and Frank [2004] produced an R² value of 0.049. Our portfolio consists of a few large capitalization stocks that follow the movements of the market closely, resulting in slightly higher adjusted R² values due to the explanatory power of the market return variable. Of primary focus in this study is the comparison between the baseline, forum, and stakeholder models and the additional refinement provided by the stakeholder analyses in the forums. Information extracted at the forum

Table XII. Stock Return Prediction Results

Model	AMZN		BA		F		HPQ		INTC		WMT		Portfolio Avg.	
	MAE	Acc %	MAE	Acc %	MAE	Acc %	MAE	Acc %	MAE	Acc %	MAE	Acc %	MAE	Acc %
Baseline	0.019	41.9	0.012	48.4	0.015	54.8	0.018	48.4	0.013	38.7	0.006	41.9	0.014	45.7
Forum	0.019	48.4	0.017	61.3	0.015	64.5	0.025	54.8	0.014	41.9	0.008	54.8	0.016	54.3**
Stakeholder	0.022	67.7*†	0.023	74.2*	0.017	67.7	0.021	51.6	0.147	61.3*	0.007	77.4***†	0.039	66.7***††

Pairwise t-tests: improvement over baseline at * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; improvement over forum at † $p < 0.1$, †† $p < 0.05$.

Table XIII. Ending Investment Values After Stock Trading

Trading Strategy	AMZN	BA	F	HPQ	INTC	WMT	Portfolio Avg.
Buy-and-Hold	\$11,400	\$10,550	\$11,690	\$12,039	\$9,828	\$9,375	\$10,814
Baseline	\$9,855	\$9,868	\$10,199	\$11,484	\$9,195	\$9,406	\$10,001
Forum	\$10,401	\$10,945	\$12,003	\$11,396	\$9,257	\$9,759	\$10,627
Stakeholder	\$12,189	\$11,540	\$12,795	\$12,154	\$11,277	\$10,685	\$11,773

level was marginally useful in explaining firm stock return, and four of the six forum models improved slightly upon their baselines. However, substantial improvements were observed in the results of the stakeholder models, achieving the highest adjusted R^2 values for each of the six firms. Information extracted at the stakeholder group level, evaluating their distinctive perspectives on each firm, provided powerful indications of subsequent stock returns.

5.3. Stock Return Prediction and Simulated Trading

The stakeholder analyses performed by the SA were also evaluated by applying the measures extracted from the forums in the prediction of daily firm stock return for 31 trading days. To directly link the stock return prediction to the regression analyses, the same models and approaches were utilized. True prediction of firm stock return was performed using only past information. The results of the baseline, forum, and stakeholder models over the 31 trading days of prediction for two evaluation criteria are presented in Table XII. The MAE remained generally consistent across model performances for each of the firms. However, substantial improvements were observed in the directional accuracy of the forum and stakeholder models. Including information at the forum level of analysis yielded on average an additional 8.6% in directional accuracy. Consistent with the results of the regression analyses, information at the stakeholder group levels provided powerful indications of subsequent firm stock returns, with statistically significant improvements in directional accuracy over the baseline models for four firms and forum models for two firms, according to pairwise t-tests. Overall, the stakeholder models correctly predicted the direction of firm stock returns for 66.7% of trading days. Performing stakeholder analysis using the SA and applying stakeholder-group-level measures improved upon the accuracy of the baseline and forum models by an average of 21% and 12.4%, respectively.

The daily stock return predictions of the baseline, forum, and stakeholder models were also applied in the simulated trading of each of the firm stocks. Over the 31 trading days, firm stocks were bought or shorted before the start of each day based on the predictions of each of the models. Trading based on the model predictions was also compared to a buy-and-hold strategy, where firm stocks were purchased at the beginning of the 31 trading days and held. Trading began with an initial investment of \$10,000 in each stock. The ending investment values for each approach are presented in Table XIII. Four of the six stocks earned positive returns over the period, and holding the portfolio yielded an impressive 8% return during the 31 trading days. These large

returns were difficult to improve upon through daily trading. Trading based on the predictions of the baseline models only broke even over the period across the portfolio. The forum models performed better, with an average ending investment of \$10,627. However, applying the predictions of the stakeholder models outperformed all other approaches for each firm, producing an average ending portfolio value of \$11,773, an impressive 17% return in only 31 trading days. The stakeholder models yielded absolute increases in return of 9% over the buy-and-hold strategy, and 17% and 11% over the baseline and forum models, respectively.

6. CONCLUSIONS

In this study, we present stakeholder analyses of firm-related web forums. A stakeholder analysis identifies distinctive groups of forum participants with shared characteristics expressed in discussion and evaluates their specific opinions and interests in the firm. To perform the stakeholder analyses, the SA for firm-related web forums was developed following the design science paradigm of information systems research and the guidance of two kernel theories, the stakeholder theory of the firm [Donaldson and Preston 1995] and the systemic functional linguistic theory [Halliday 2004]. We describe and justify the design of the SA and evaluate its performance in the intended application to demonstrate its effectiveness, but also recognize that design alternatives exist for various aspects of the system, and the design process is an inherently iterative search [Hevner et al. 2004]. We encourage future research to continue the design science process, building upon our design or proposing alternatives, to advance this class of information systems for performing stakeholder analysis in firm-related forums.

Stakeholder analyses were performed on six major firm-related forums hosted on Yahoo Finance over a 3-month period and a total of 63,450 messages written by 6,372 unique participants. Following prior studies [Antweiler and Frank 2004; Das and Chen 2007], the relationships between measures extracted from the forums and subsequent daily firm stock returns were examined over 62 trading days using multiple linear regression models. We compared baseline models consisting of established predictors of stock return to models that included information extracted at the forum or stakeholder group levels of analysis. The regression analyses revealed statistically significant indicators of subsequent daily firm stock returns in the prior discussions of the stakeholder groups of each firm, with stakeholder-model-adjusted R^2 values reaching 0.83. Using the same models and comparisons, we also performed daily stock return prediction for 31 trading days and found that the stakeholder models correctly predicted the direction of return on 67% of trading days, generating an impressive 17% return in simulated trading of the six firm stocks. We focused on these six major firms to demonstrate stakeholder analysis over a diverse and representative sample, but recognize the portfolio and time period of analysis was limited, and the findings of this study may not generalize to all other firms or time periods. However, these evaluations demonstrated that the stakeholder analyses provided more refined assessments of each of the six firm-related forums, yielding measures at the stakeholder group level that better explain and predict subsequent firm stock returns compared to aggregate forum-level information.

6.1. Implications for Management

The findings of the study indicate that firm-related web forums may contain distinctive groups within the population of participants. Performing a stakeholder analysis to identify interest-based stakeholder groups, and evaluating their distinctive perspectives

and opinions on the firm, refines the analysis and assessment of the forum. Management should reconsider assessing firm-related forums, as well as other forms of firm-related social media, in the aggregate, but rather apply a stakeholder analysis approach to these diverse populations.

The devised SA system offers management the capability to automatically analyze and monitor firm-related forums at a large scale over time and identify and characterize interest-based stakeholder groups participating in discussion and their sentiments expressed on the firm. The business intelligence gained through application of the SA allows for real-time, targeted reorientation of stakeholder management activities to continuously improve relations with various groups. Improved understanding of stakeholder groups and their distinctive perspectives represents a significant business value with benefits throughout the firm [Barua et al. 2004].

The findings of the stock return regression analyses and prediction stress the importance of effective stakeholder management. For each of the six firms, daily stock return was indicated in advance by the discussions of various stakeholder groups in the forums. Stakeholder analyses could provide firm management with explanations for fluctuations in past stock return and the knowledge required to influence future returns by satisfying specific stakeholder group interests.

6.2. Implications for Research

Prior studies on firm-related forums and stock behavior specifically targeting a uniform population of investors may have actually focused on a fraction of participants and discussions. The stakeholder analyses revealed diverse stakeholder groups participating in each firm-related forum, and accounting for their distinctive perspectives on the firms improved our ability to explain and predict subsequent daily firm stock returns. Generally, research on firm-related social media and indicators of firm performance may refine the extracted information and improve the explanatory power of these sources by applying a stakeholder analysis approach to identify interest-based stakeholder groups, even when one functional role-based stakeholder group (like consumers or investors) is presumed to be generating the social media.

This research provides a form of empirical support for the stakeholder theory of the firm not previously established in the literature. We also design a stakeholder feature representation corresponding to the SFLT to comprehensively characterize participation in forums and serve as a basis for the stakeholder analyses. The stakeholder feature representation presented in this study could be adapted to suit other genres of social media, continuing the design process.

6.3. Future Research Directions

We will engage in future research and iterations in the design process to improve and extend the approaches to stakeholder analysis and the SA presented in this study. In addition to the various design aspects to be explored as discussed throughout the description of the system, future directions for this research include applying the SA to firm-related forums hosted on other major web portals and smaller independent forums. We will expand the firm portfolio, particularly to include smaller firms, and lengthen the time span for the analysis. We will also examine the performance of the SA when deployed for stakeholder analysis and stock return prediction at more frequent intervals of time. The SA will also be adapted to analyze, monitor, and integrate information from multiple firm-related social media, including blogs, reviews, and Twitter messages, to form a more comprehensive social media analytics platform.

APPENDIX A: SENTIMENT CLASS-LEVEL PERFORMANCE OF SA IN MESSAGE SENTIMENT CLASSIFICATION

	Positive Class Messages			Negative Class Messages			Neutral Class Messages		
	Precision	Recall	F-Measure	Precision	Recall	F-Measure	Precision	Recall	F-Measure
AMZN	0.61	0.87	0.71	1.0	0.5	0.67	0.76	0.87	0.81
BA	0.45	0.88	0.60	0.93	0.64	0.76	0.79	0.78	0.78
F	0.36	0.73	0.49	1.0	0.55	0.71	0.75	0.87	0.80
HPQ	0.68	0.85	0.76	1.0	0.64	0.78	0.69	0.94	0.79
INTC	0.72	0.88	0.79	1.0	0.73	0.85	0.74	0.90	0.81
WMT	0.50	0.75	0.60	0.89	0.60	0.71	0.75	0.86	0.80
Average	0.55	0.83	0.66	0.97	0.61	0.75	0.75	0.87	0.80

APPENDIX B: STAKEHOLDER GROUP KEY PHRASE EXTRACTION RESULTS

Amazon	Avg. Prob.	Keywords	Key Phrases
Stakeholder Group 1 “firm analysis”	0.15	sales price company tax growth model	long term business model sales tax holiday sales fiscal cliff rev growth
Stakeholder Group 2 “stock traders”	0.32	buy pe short stock sell crash	good short weak handed max pain forward pe short term big drop
Stakeholder Group 3 “stock market analysis”	0.12	stock market shares money world move	million shares federal reserve stock market makes money stock trading year end
Stakeholder Group 4 “financial analysis”	0.09	billion tax current cash flow capital	gross profit cash flow market cap gross margin operating cash bond market
Stakeholder Group 5 “stock analysis”	0.25	stock short earnings sell buy pe	revenue growth buy puts high pe put money earnings report rock solid
Stakeholder Group 6 “information exchange”	0.06	report thanks great post look show	good point good job bad news time buy good luck sell short

Boeing	Avg. Prob.	Keywords	Key Phrases
Stakeholder Group 1 “government - R”	0.14	republican gun spending fuel security government	spending cut quick fix assault weapons social security gun lobby obama cut
Stakeholder Group 2 “development issues”	0.31	battery problems planes aircraft lithium design	lithium batteries real problem carbon fiber lithium ion battery problem electrical system
Stakeholder Group 3 “government - D”	0.07	obama people american government president country	obama democrats mr obama pay benefits federal reserve tax increases financial crisis
Stakeholder Group 4 “information exchange”	0.07	good read information hear article report	stock market good thing good post good news billion dollar bad news
Stakeholder Group 5 “aircraft”	0.35	planes company batteries flying problem risk	made money real world electrical problems long term fuel leaks battery fire

Ford	Avg. Prob.	Keywords	Key Phrases
Stakeholder Group 1 “election and labor”	0.26	america gop people election business liberals	percent americans health insurance federal government labor force united states employment based
Stakeholder Group 2 “government - populace”	0.12	people think years hard money system	air force president years obama won people want middle class real world
Stakeholder Group 3 “government - fiscal”	0.11	tax obama money republican taxes security	tax cuts jobs created social security global warming fiscal cliff medical care

Ford	Avg. Prob.	Keywords	Key Phrases
Stakeholder Group 4 “tea party”	0.20	people obama republican party country europe	real father hate america republican party tea party tax increase barack obama
Stakeholder Group 5 “information exchange”	0.05	yahoo thanks check good message report	finance yahoo message board tax cuts stock market executive orders fox news
Stakeholder Group 6 “stock analysis”	0.22	buy stock price time short share	big time price target market share dividend increase private sector stock price

Hewlett-Packard	Avg. Prob.	Keywords	Key Phrases
Stakeholder Group 1 “firm analysis”	0.30	buy company meg citizens money stock	long term buy shares balance sheet due diligence stock price carly fiorina
Stakeholder Group 2 “small talk”	0.38	good buy run sell stock lol	good news ha ha good luck common sense make money buy buy
Stakeholder Group 3 “market analysis”	0.16	people company money business billion debt	tax rates pc sales people money global warming thank god pay taxes
Stakeholder Group 4 “exercise stock option”	0.07	direct dec share tax state economy	share dec direct option option exercise exercise share open market direct disposition
Stakeholder Group 5 “information exchange”	0.07	thanks report great good look read	look good profit loss dow nas crude gold euro dollar autonomy deal

Intel	Avg. Prob.	Keywords	Key Phrases
Stakeholder Group 1 “stock analysis”	0.34	buy stock dividend price market shares	good buy good time buy back buy buy stock price collect dividend
Stakeholder Group 2 “information exchange”	0.08	report looking thanks great good article	makes sense good job make money great news good things informational article
Stakeholder Group 3 “message board”	0.17	price yahoo stock market message believe	finance yahoo message board wall street stock price buy sell seekingalpha com
Stakeholder Group 4 “products and manufacturing”	0.06	fab market apple pc motorola designs	big money total cost market share motorola mobility personal computers fab process
Stakeholder Group 5 “small talk”	0.25	good hope great lol post thanks	ha ha big deal clover trail good luck make sense great news

Wal-Mart	Avg. Prob.	Keywords	Key Phrases
Stakeholder Group 1 “information exchange”	0.50	good thanks great look news sales	stock market made money great job post topic press release good news
Stakeholder Group 2 “stores and expansion”	0.13	open stores workers state store union	stores open private sector china india walmart stores unemployment benefits nation largest
Stakeholder Group 3 “labor and union”	0.34	union workers pay money work days	food stamp union workers minimum wage work thanksgiving health care pay workers

APPENDIX C: CORRELATION ANALYSIS

Correlations (n = 62)		AMZN	BA	F	HPQ	INTC	WMT
		Return _t	Return _t	Return _t	Return _t	Return _t	Return _t
Stock	DJI Return _t	0.411***	0.523***	0.428***	0.450***	0.501***	0.446***
	DJI Return _{t-1}	0.191	-0.044	0.097	-0.137	-0.129	0.071
	Return _{t-1}	-0.083	-0.062	0.183	-0.205	-0.035	0.043
	Volatility _{t-1}	0.011	0.122	0.034	0.248*	-0.381***	-0.285**
	Volume _{t-1}	-0.036	0.119	0.118	0.153	-0.172	-0.084
Forum	Num. Messages _{t-1}	-0.100	-0.073	-0.310**	-0.011	-0.083	0.087
	Avg. Length _{t-1}	-0.058	-0.149	0.398***	-0.071	0.158	-0.149
	Std. Dev. Length _{t-1}	-0.025	-0.099	0.315**	-0.061	0.125	-0.016
	Avg. Subjectivity _{t-1}	-0.078	-0.164	0.413***	-0.129	0.174	-0.192
	Std. Dev. Subjectivity _{t-1}	-0.112	-0.171	0.268**	-0.049	0.113	-0.092
	Avg. Sentiment _{t-1}	-0.037	0.074	-0.212	0.040	0.042	-0.129
	Std. Dev. Sentiment _{t-1}	-0.176	-0.126	0.245*	0.047	0.027	-0.169

*p < 0.1; **p < 0.05; ***p < 0.01; ****p < 0.001.

APPENDIX D: BASELINE STOCK RETURN REGRESSION ANALYSIS RESULTS

Baseline Stock Return Regression		AMZN	BA	F	HPQ	INTC	WMT
		Return _t	Return _t	Return _t	Return _t	Return _t	Return _t
	Intercept	-0.086 (-0.649)	-0.015 (-0.294)	-0.054 (-0.563)	0.052 (0.302)	0.073 (0.073)	-0.012 (-0.207)
Stock	MKT Return _t	1.023*** (3.464)	0.953*** (4.749)	0.896*** (3.466)	1.496*** (3.473)	1.049*** (4.567)	0.520*** (3.191)
	MKT Return _{t-1}	0.613* (1.971)	-0.073 (-0.303)	0.110 (0.388)	-0.344 (-0.725)	-0.270 (-1.003)	0.165 (0.869)
	Return _{t-1}	-0.197 (-1.513)	0.003 (0.027)	0.084 (0.738)	-0.188 (-1.466)	-0.047 (-0.378)	-0.084 (-0.569)
	Volatility _{t-1}	-0.239 (-0.686)	0.262 (1.081)	0.027 (0.110)	0.405 (1.163)	-0.718*** (-2.793)	-0.496* (-1.756)
	Volume _{t-1}	0.014 (0.667)	0.001 (0.205)	0.007 (0.549)	-0.008 (-0.347)	-0.007 (-0.743)	0.002 (0.279)
R ²		0.232	0.302	0.214	0.277	0.391	0.246
Adjusted R ²		0.164	0.240	0.144	0.213	0.336	0.179

*p < 0.1; **p < 0.05; ***p < 0.01; ****p < 0.001.

APPENDIX E: STAKEHOLDER STOCK RETURN REGRESSION ANALYSIS RESULTS

AMZN		Return _t
	Intercept	0.077 (0.905)
Stock	DJI Return _t	1.553 (7.696)***
	DJI Return _{t-1}	0.310 (1.730)*
	Return _{t-1}	0.029 (0.425)
	Volatility _{t-1}	-0.698 (-2.924)***
	Volume _{t-1}	-0.003 (-0.202)
Stakeholder Group 1 “firm analysis” Avg. Prob.: 0.15	Num. Messages _{t-1}	-0.010 (-3.005)***
	Avg. Subjectivity _{t-1}	-0.002 (-1.614)
	Std. Dev. Subjectivity _{t-1}	0.000 (1.639)
	Avg. Sentiment _{t-1}	0.008 (3.570)**
	Std. Dev. Sentiment _{t-1}	-0.002 (-3.208)***
Stakeholder Group 2 “stock traders” Avg. Prob.: 0.32	Num. Messages _{t-1}	0.007 (2.363)**
	Std. Dev. Length _{t-1}	0.000 (3.707)***
	Avg. Subjectivity _{t-1}	0.003 (1.370)
	Std. Dev. Subjectivity _{t-1}	-0.001 (-1.023)
Stakeholder Group 3 “stock market analysis” Avg. Prob.: 0.12	Num. Messages _{t-1}	-0.004 (-1.655)
	Avg. Length _{t-1}	-0.000 (-1.646)
	Avg. Sentiment _{t-1}	0.002 (1.223)
	Std. Dev. Sentiment _{t-1}	0.003 (3.419)**
Stakeholder Group 4 “financial analysis” Avg. Prob.: 0.09	Avg. Length _{t-1}	-0.001 (-5.185)***
	Avg. Subjectivity _{t-1}	0.004 (1.116)
	Std. Dev. Subjectivity _{t-1}	0.001 (3.852)***
	Std. Dev. Sentiment _{t-1}	-0.000 (-1.431)
Stakeholder Group 5 “stock analysis” Avg. Prob.: 0.25	Avg. Length _{t-1}	0.000 (4.066)***
	Std. Dev. Length _{t-1}	-0.000 (-4.534)***
	Std. Dev. Subjectivity _{t-1}	0.001 (4.399)***
	Std. Dev. Sentiment _{t-1}	-0.001 (-4.622)***
Stakeholder Group 6 “information exchange” Avg. Prob.: 0.06	Num. Messages _{t-1}	-0.019 (-5.724)***
	Avg. Length _{t-1}	0.001 (2.289)**
	Std. Dev. Length _{t-1}	-0.001 (-4.251)***
	Avg. Subjectivity _{t-1}	0.763 (7.760)***
	Std. Dev. Subjectivity _{t-1}	-0.462 (-8.015)***
	Avg. Sentiment _{t-1}	-0.399 (-7.985)***
	Std. Dev. Sentiment _{t-1}	0.230 (8.836)***
R ²		0.920
Adjusted R ²		0.826

*p < 0.1; **p < 0.05; ***p < 0.01; ****p < 0.001.

BA		Return _t
	Intercept	-0.001 (-0.023)
Stock	DJI Return _t	1.238 (6.292)***
	DJI Return _{t-1}	-0.401 (-1.639)
	Return _{t-1}	0.000 (0.003)
	Volatility _{t-1}	0.155 (0.698)
	Volume _{t-1}	0.001 (0.075)
Stakeholder Group 1 “government - R” Avg. Prob.: 0.14	Avg. Sentiment _{t-1}	0.006 (2.896)***

BA		Return _t
Stakeholder Group 2 “development issues” Avg. Prob.: 0.31	Avg. Length _{t-1}	-0.000 (-2.668)**
	Avg. Sentiment _{t-1}	-0.004 (-2.666)**
Stakeholder Group 3 “government - D” Avg. Prob.: 0.07	Std. Dev. Sentiment _{t-1}	-0.000 (-2.350)**
Stakeholder Group 5 “aircraft” Avg. Prob.: 0.35	Num. Messages _{t-1}	0.002 (1.555)
R ²		0.491
Adjusted R ²		0.391

*p < 0.1; **p < 0.05; ***p < 0.01; ****p < 0.001.

F		Return _t
Stock	Intercept	-0.087 (-1.266)
	DJI Return _t	1.257 (7.253)****
	DJI Return _{t-1}	0.369 (2.194)**
	Return _{t-1}	0.040 (0.592)
	Volatility _{t-1}	-0.199 (-1.072)
	Volume _{t-1}	0.015 (1.595)
Stakeholder Group 1 “election and labor” Avg. Prob.: 0.26	Num. Messages _{t-1}	-0.004 (-1.623)
	Avg. Length _{t-1}	-0.000 (-1.999)*
	Std. Dev. Length _{t-1}	0.000 (1.894)*
	Avg. Sentiment _{t-1}	-0.005 (-1.271)
Stakeholder Group 2 “government - populace” Avg. Prob.: 0.12	Std. Dev. Length _{t-1}	0.000 (3.950)****
	Avg. Subjectivity _{t-1}	0.001 (1.537)
	Std. Dev. Subjectivity _{t-1}	-0.001 (-5.059)****
	Avg. Sentiment _{t-1}	-0.004 (-3.498)**
Stakeholder Group 3 “government - fiscal” Avg. Prob.: 0.11	Std. Dev. Sentiment _{t-1}	-0.001 (-2.608)**
	Num. Messages _{t-1}	-0.008 (-2.504)**
	Avg. Length _{t-1}	-0.000 (-2.114)**
	Std. Dev. Length _{t-1}	0.000 (1.959)*
Stakeholder Group 4 “tea party” Avg. Prob.: 0.20	Avg. Sentiment _{t-1}	-0.002 (-1.243)
	Num. Messages _{t-1}	0.020 (4.317)****
	Std. Dev. Length _{t-1}	0.000 (1.177)
Stakeholder Group 5 “information exchange” Avg. Prob.: 0.05	Avg. Subjectivity _{t-1}	-0.005 (-2.347)**
	Num. Messages _{t-1}	-0.004 (-1.985)*
	Avg. Length _{t-1}	0.001 (4.602)****
	Avg. Subjectivity _{t-1}	0.013 (3.216)**
	Std. Dev. Subjectivity _{t-1}	0.013 (8.575)****
Stakeholder Group 6 “stock analysis” Avg. Prob.: 0.22	Std. Dev. Sentiment _{t-1}	-0.013 (-8.924)****
	Num. Messages _{t-1}	-0.007 (-2.264)**
	Avg. Length _{t-1}	-0.001 (-4.030)****
	Std. Dev. Subjectivity _{t-1}	0.002 (3.018)**
	Avg. Sentiment _{t-1}	0.004 (1.783)*
Std. Dev. Sentiment _{t-1}		0.003 (2.917)**
R ²		0.899
Adjusted R ²		0.795

*p < 0.1; **p < 0.05; ***p < 0.01; ****p < 0.001.

HPQ		Return _t
	Intercept	-0.080 (-0.484)
Stock	DJI Return _t	2.115 (5.463)****
	DJI Return _{t-1}	0.608 (1.511)
	Return _{t-1}	-0.247 (-2.412)**
	Volatility _{t-1}	0.150 (0.490)
	Volume _{t-1}	0.013 (0.553)
Stakeholder Group 1 “firm analysis” Avg. Prob.: 0.30	Avg. Length _{t-1}	0.000 (2.272)**
	Std. Dev. Subjectivity _{t-1}	-0.001 (-1.897)*
	Avg. Sentiment _{t-1}	-0.004 (-1.503)
	Std. Dev. Sentiment _{t-1}	0.002 (2.329)**
Stakeholder Group 2 “small talk” Avg. Prob.: 0.38	Avg. Length _{t-1}	-0.001 (-1.543)
	Std. Dev. Length _{t-1}	0.000 (2.146)**
Stakeholder Group 3 “market analysis” Avg. Prob.: 0.16	Num. Messages _{t-1}	-0.011 (-2.726)**
	Avg. Length _{t-1}	0.001 (2.682)**
	Std. Dev. Length _{t-1}	-0.000 (-2.609)**
	Avg. Subjectivity _{t-1}	-0.018 (-3.122)***
	Std. Dev. Subjectivity _{t-1}	0.000 (1.431)
	Std. Dev. Sentiment _{t-1}	0.002 (2.782)***
Stakeholder Group 4 “exercise stock option” Avg. Prob.: 0.07	Avg. Length _{t-1}	0.000 (1.457)
	Std. Dev. Length _{t-1}	-0.000 (-2.516)**
	Avg. Subjectivity _{t-1}	-0.009 (-2.177)**
	Avg. Sentiment _{t-1}	-0.012 (-2.944)***
	Std. Dev. Sentiment _{t-1}	0.002 (4.714)****
Stakeholder Group 5 “info exchange” Avg. Prob.: 0.07	Avg. Length _{t-1}	0.000 (1.280)
	Std. Dev. Length _{t-1}	-0.000 (-4.230)****
	Avg. Subjectivity _{t-1}	-0.013 (-2.925)***
	Std. Dev. Subjectivity _{t-1}	0.001 (4.716)****
	Avg. Sentiment _{t-1}	0.009 (2.498)**
R ²		0.785
Adjusted R ²		0.614

*p < 0.1; **p < 0.05; ***p < 0.01; ****p < 0.001.

INTC		Return _t
	Intercept	0.080 (1.003)
Stock	DJI Return _t	1.069 (4.839)****
	DJI Return _{t-1}	-0.289 (-1.154)
	Return _{t-1}	-0.051 (-0.422)
	Volatility _{t-1}	-0.501 (-1.980)*
	Volume _{t-1}	-0.010 (-0.989)
Stakeholder Group 1 “stock analysis” Avg. Prob.: 0.34	Num. Messages _{t-1}	-0.007 (-2.769)**
	Avg. Length _{t-1}	0.001 (2.227)**
	Avg. Subjectivity _{t-1}	-0.005 (-2.238)**
	Std. Dev. Subjectivity _{t-1}	-0.000 (-1.265)
	Avg. Sentiment _{t-1}	-0.007 (-3.229)***
Stakeholder Group 2 “information exchange” Avg. Prob.: 0.08	Avg. Length _{t-1}	-0.002 (-2.426)**
	Std. Dev. Subjectivity _{t-1}	0.009 (2.196)**

INTC		Return _t
Stakeholder Group 3 “message board” Avg. Prob.: 0.17	Num. Messages _{t-1}	0.006 (2.291)**
	Avg. Length _{t-1}	0.000 (1.378)
	Std. Dev. Subjectivity _{t-1}	-0.000 (-2.995)***
Stakeholder Group 4 “products and manufacturing” Avg. Prob.: 0.06	Std. Dev. Length _{t-1}	0.000 (1.503)
Stakeholder Group 5 “small talk” Avg. Prob.: 0.25	Std. Dev. Length _{t-1}	-0.000 (-2.638)**
	Avg. Subjectivity _{t-1}	0.004 (2.214)**
R ²		0.669
Adjusted R ²		0.531

*p < 0.1, **p < 0.05; ***p < 0.01; ****p < 0.001.

WMT		Return _t
Stock	Intercept	0.010 (0.184)
	DJI Return _t	0.880 (5.084)****
	DJI Return _{t-1}	0.012 (0.069)
	Return _{t-1}	0.062 (0.426)
	Volatility _{t-1}	0.051 (0.184)
	Volume _{t-1}	0.001 (0.134)
Stakeholder Group 1 “information exchange” Avg. Prob.: 0.50	Num. Messages _{t-1}	0.006 (2.767)***
	Std. Dev. Subjectivity _{t-1}	0.001 (1.728)*
	Std. Dev. Sentiment _{t-1}	-0.002 (-2.419)**
Stakeholder Group 2 “stores and expansion” Avg. Prob.: 0.13	Avg. Length _{t-1}	-0.000 (-2.722)***
	Std. Dev. Length _{t-1}	0.000 (1.856)*
	Avg. Sentiment _{t-1}	-0.005 (-1.708)*
	Std. Dev. Sentiment _{t-1}	-0.000 (-1.474)
Stakeholder Group 3 “labor and union” Avg. Prob.: 0.34	Num. Messages _{t-1}	-0.005 (-2.770)***
	Std. Dev. Length _{t-1}	-0.000 (-2.451)**
	Avg. Subjectivity _{t-1}	-0.001 (-1.377)
	Std. Dev. Subjectivity _{t-1}	0.000 (3.624)****
	Avg. Sentiment _{t-1}	-0.002 (-2.343)**
	Std. Dev. Sentiment _{t-1}	0.000 (2.424)**
R ²		0.575
Adjusted R ²		0.397

*p < 0.1, **p < 0.05; ***p < 0.01; ****p < 0.001.

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